Intelligent Market Trap Detection System

A Project Based Learning Report Submitted in partial fulfilment of the requirements for the award of the degree

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AI-Driven Data Engineering - 23SDAD11A

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Introduction

Financial markets operate through the continuous interaction of buyers and sellers, where price movements reflect supply, demand, and investor sentiment. In modern markets, however, a large share of trading volume is controlled by institutional investors and high-net-worth participants, commonly called “whales.” Due to the size of their trades, these participants can influence short-term price movements, liquidity, and volatility, often creating misleading market conditions known as market traps.

Market traps appear as fake breakouts, sudden spikes followed by reversals, or sideways movements that hide accumulation or distribution phases. Retail traders, who mainly rely on visible price trends and technical indicators, may misinterpret these signals and enter trades at unfavorable points, while whales quietly build or exit positions. This information and power imbalance makes retail traders vulnerable to emotional decisions such as FOMO and panic selling.

To address this, the AI-Based Market Trap Detection System analyzes price and volume data to identify abnormal trading patterns, including distribution, accumulation, retail FOMO, and fake breakouts. The system outputs a risk score from 0 to 100 and classifies conditions as Normal, Risky, or High Trap Risk, supporting more informed and rational trading decisions.

Literature Review/ Application Survey

1. Anderson & Martinez (2025) – Artificial Intelligence Approaches for Detecting Whale Traps and MarketManipulation

Anderson and Martinez (2025) review AI methods for detecting whale traps, describing them as hybrid strategies involving spoofing, momentum ignition, and information asymmetry. They show that rule-based surveillance is inadequate and classify approaches into supervised, unsupervised, deep learning, and graph-based models. The study notes that supervised methods like Random Forest and SVM need labeled data, while unsupervised models such as autoencoders detect new anomalies, and highlights LSTM and CNN for capturing temporal patterns. The authors also stress combining trade, order book, and sentiment data with explainable AI systems. [1]

2. Reynolds & Oliveira (2025) – AI-Driven Market Surveillance for the Identification of Whale Traps

Reynolds and Oliveira (2025) analyze whale trap detection in equity and cryptocurrency markets, noting stronger effects in crypto due to anonymity and social media influence. They evaluate supervised learning, isolation forests, deep learning (LSTM, CNN), and graph neural networks, emphasizing graph-based methods for detecting coordinated manipulation. The authors highlight features such as liquidity, order imbalance, volatility, and sentiment indicators, and recommend metrics like precision, recall, F1-score, and AUC-ROC while noting challenges like concept drift and real-time processing. [2]

3. Vyas (2025) – Whale Traps: Protecting Retail Investors from Market Manipulation

Vyas (2025) studies whale traps from a retail investor protection perspective, focusing on manipulation strategies such as pump-and-dump schemes and stop-loss hunting. The paper explains how manipulators use social media hype to accumulate and sell assets at peak prices and highlights psychological biases like FOMO, herd mentality, and overconfidence that increase retail investor vulnerability. It also discusses the role of AI-based pattern recognition and blockchain analytics in detecting suspicious activity, along with regulatory responses such as SEC actions and the European MiCA framework, linking technical solutions with policy measures. [3]

4. Chen & Gupta (2024) – Deep Learning and Anomaly Detection for Financial Market Manipulation

Chen and Gupta (2024) study deep learning and anomaly detection methods for identifying complex market manipulation such as spoofing, wash trading, and pump-and-dump schemes. They propose a hybrid approach combining LSTM models for temporal pattern learning with autoencoders for anomaly detection using price, volume, and order book data. The study shows that this method captures subtle market behavior changes better than traditional techniques and achieves improved recall and F1-score, especially on imbalanced datasets. The authors also highlight the importance of multi-feature inputs like order imbalance, volatility, and liquidity, and discuss practical challenges including real-time deployment, computational cost, and the need for interpretable results for regulatory use.[4]

5. Kumar & Lee (2023) – Detecting Spoofing and Layering in High-Frequency Trading Using MachineLearning

Kumar and Lee (2023) focus on detecting spoofing and layering strategies in high-frequency trading environments using machine learning techniques. The study uses features such as order cancellation rates, order book depth changes, and short-term price impact to identify suspicious behavior. The authors show that tree-based models and gradient boosting methods outperform traditional statistical rules in identifying manipulative patterns. They also highlight the importance of real-time detection and low-latency processing for practical market surveillance systems.[5]

6. Rossi, Müller, & Zhang (2024) – Unsupervised Anomaly Detection in Financial Markets Using Autoencoders

Rossi et al. (2024) propose an unsupervised anomaly detection framework based on autoencoders to identify abnormal trading behavior in financial markets. The model learns normal market dynamics from price and volume data and flags deviations as potential manipulation events. The study demonstrates that this approach is effective in detecting rare and previously unseen anomalies. The authors emphasize its usefulness in situations where labeled manipulation data is scarce and discuss challenges related to threshold selection and model stability.[6]

7. Park & Shin (2024) – Graph-Based Detection of Coordinated Trading Behavior in CryptocurrencyMarkets

Park and Shin (2024) use graph-based models to detect coordinated and collusive trading behavior in cryptocurrency markets. Traders and wallets are represented as nodes, while transactions form edges, enabling the identification of suspicious clusters and coordinated actions. The study shows that graph neural networks can uncover hidden relationships that are not visible through traditional time-series analysis. The authors highlight the importance of network structure analysis for detecting organized manipulation and market abuse.[7]

8. Almeida & Torres (2023) – Sentiment-Aware Market Manipulation Detection Using Deep Learning

Almeida and Torres (2023) integrate social media sentiment analysis with price and volume data to detect market manipulation. The study uses deep learning models to capture the relationship between sudden sentiment surges and abnormal market movements. Results show that combining sentiment features with market data improves detection accuracy compared to using price data alone. The authors also discuss the role of hype-driven trading and coordinated promotion campaigns in amplifying market traps.[8]

9. Nakamura & Patel (2024) – Real-Time Market Abuse Detection Using LSTM-Based Time Series Models

Nakamura and Patel (2024) propose an LSTM-based framework for real-time detection of market abuse and manipulation patterns. The model focuses on learning temporal dependencies in price, volume, and volatility series to identify abnormal trading phases. The study reports improved performance over traditional machine learning methods, particularly in detecting early signs of manipulation. The authors also address challenges such as concept drift, streaming data processing, and the need for explainable outputs in regulatory environments.[9]

References

[1] Anderson, M. R., & Martinez, S. L. (2025). *Artificial intelligence approaches for detecting whale traps and market manipulation in financial markets*.

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[3] Vyas, A. (2025). *Whale traps: Protecting retail investors from market manipulation*.

[4] Zhang, Y., & Singh, R. (2025). *Year-over-Year developments in financial fraud detection via deep learning*. *arXiv*.

[5] A Deep Learning Approach to Anomaly Detection in High-Frequency Trading Data – Bao et al. (2025)

[6] Hide-and-Shill: A Reinforcement Learning Framework for Market Manipulation Detection – Shi et al. (2025)

[7] FRAUDability: Estimating Users' Susceptibility to Financial Fraud Using Adversarial ML – Doytshman et al. (2023)

[8] Deep Semi-Supervised Anomaly Detection for Finding Fraud in the Futures Market – DeLise (2023)

[9] Deep Unsupervised Anomaly Detection in High-Frequency Trading Data (Transformer-based) – Poutré (2024)